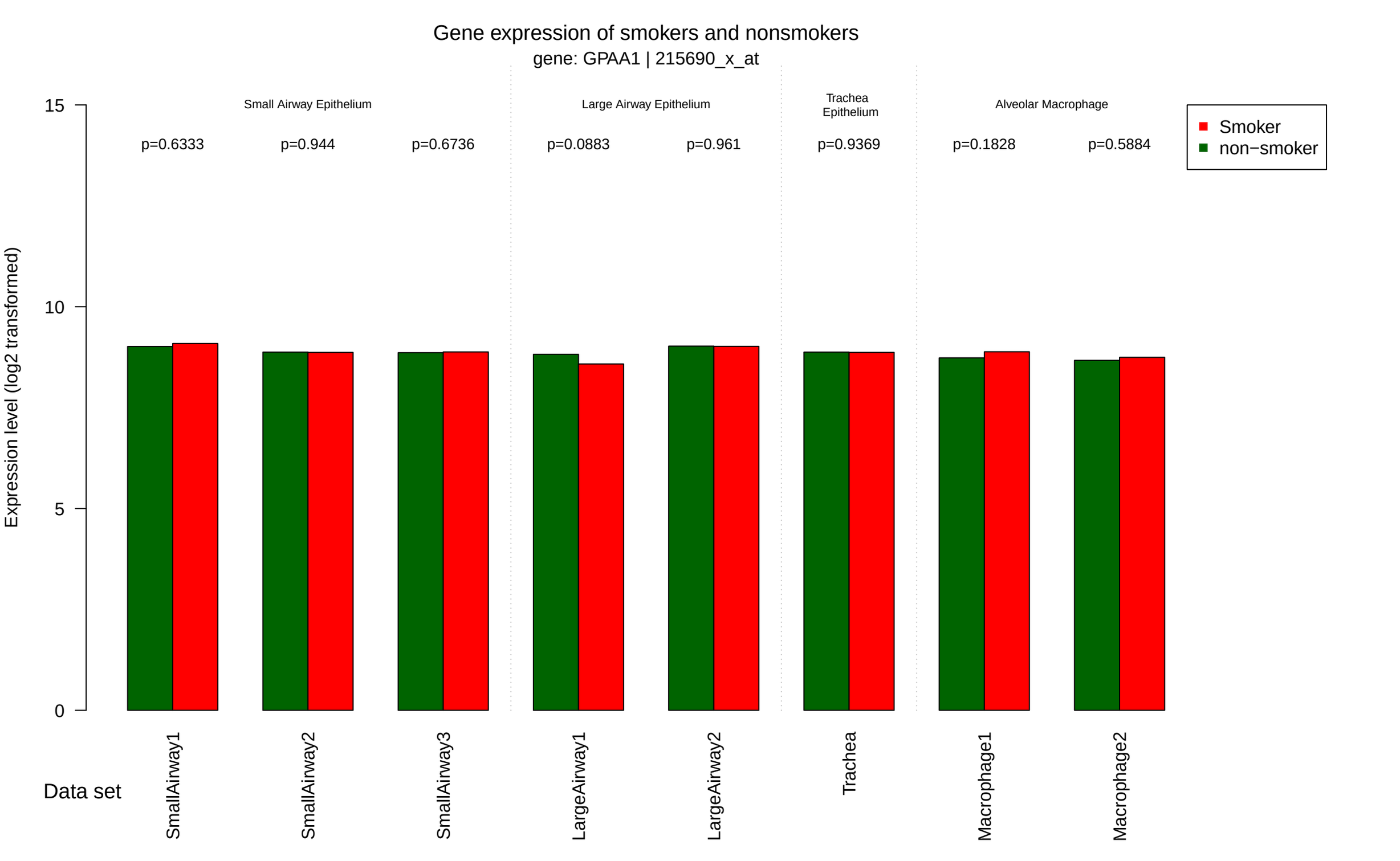
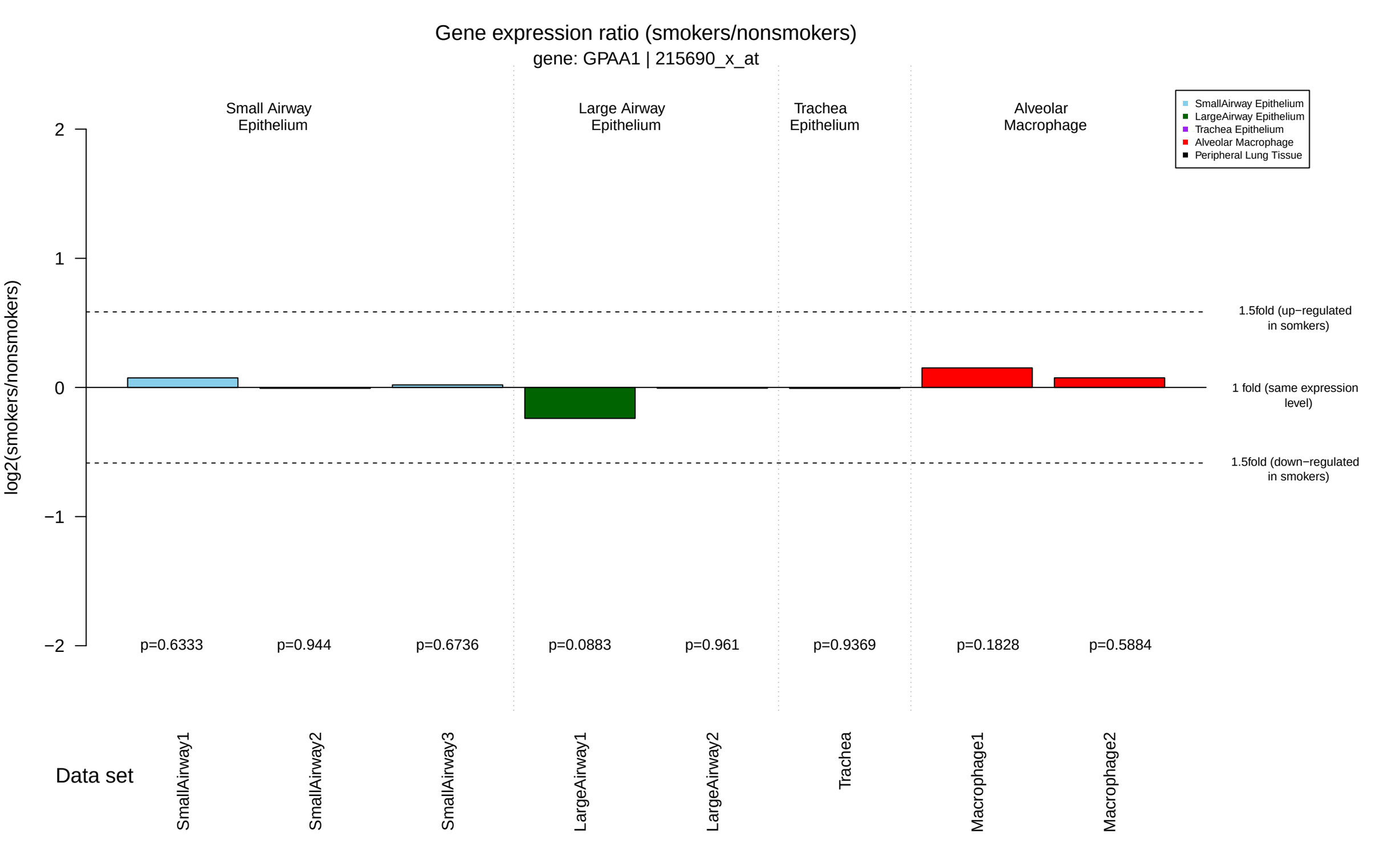
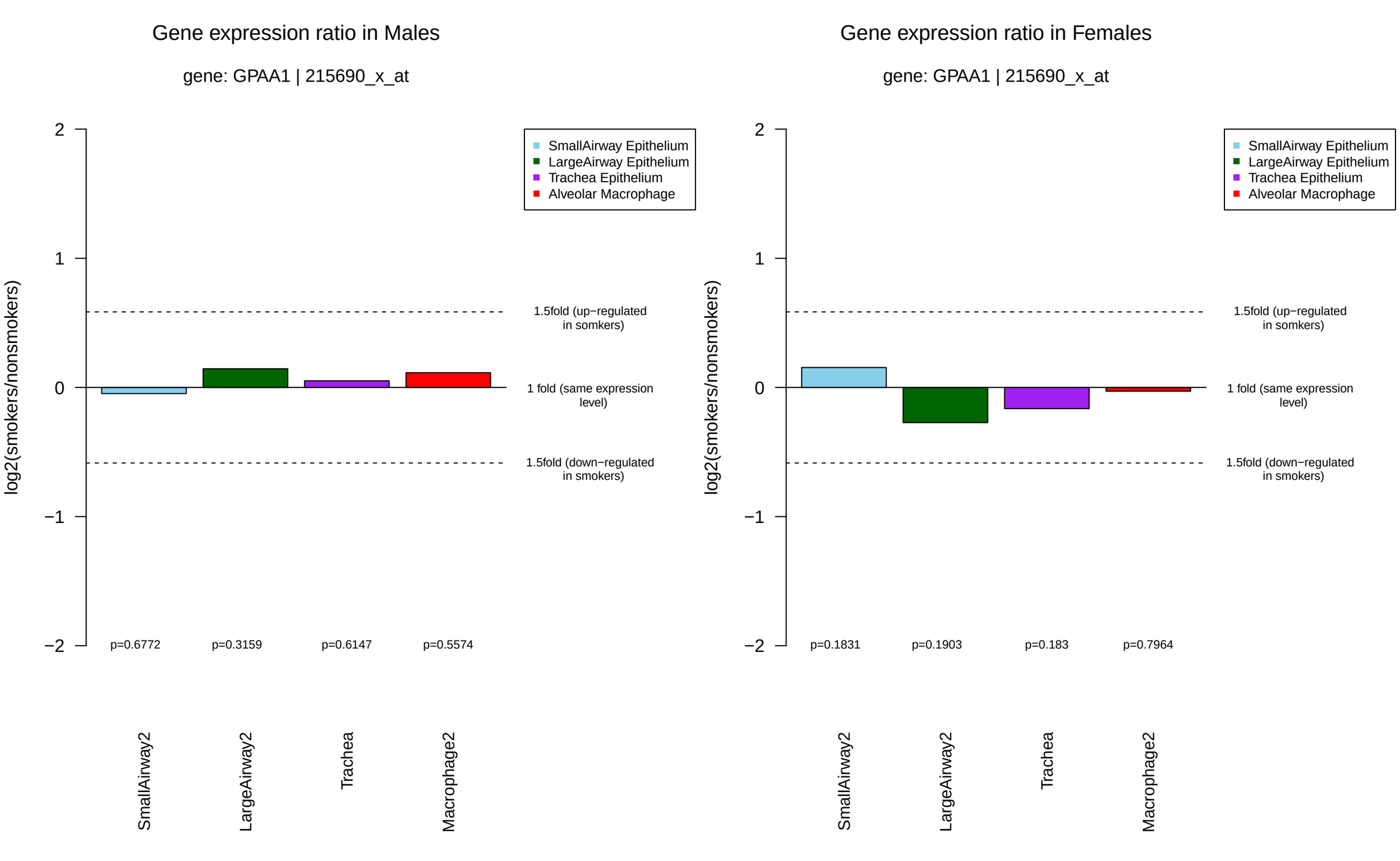
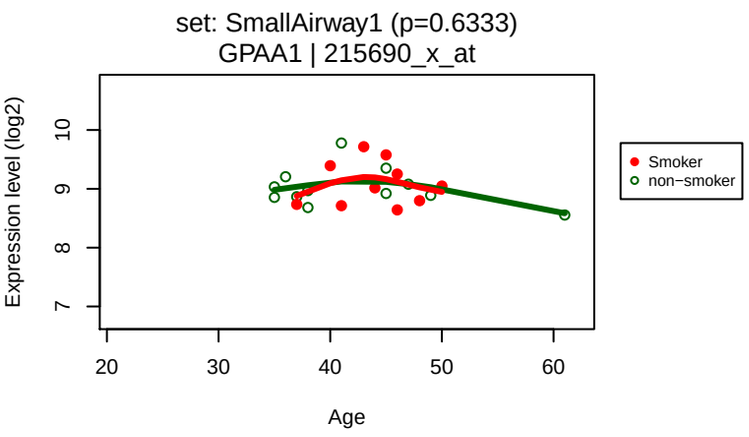
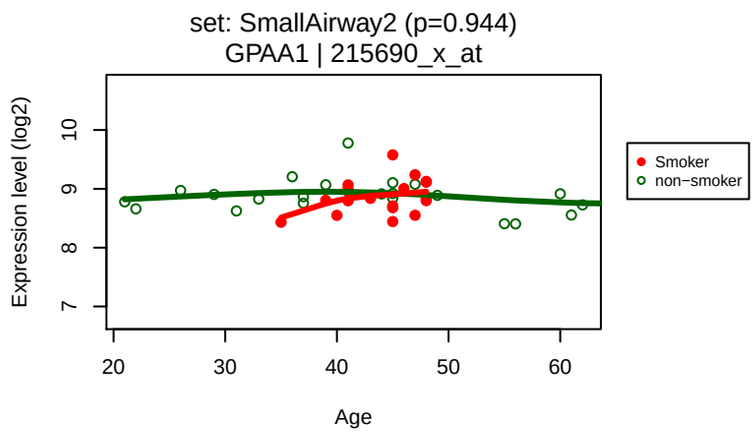
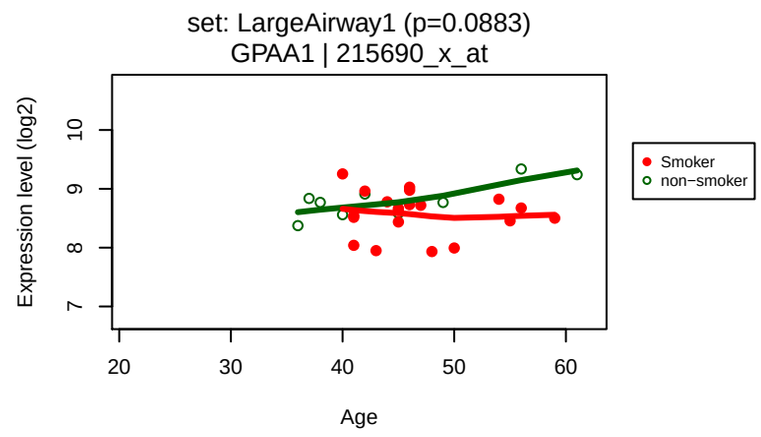

Gene expression of smokers and nonsmokers
gene: GPAA1 | 215690_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6333
p=0.944
p=0.6736
p=0.0883
p=0.961
p=0.9369
p=0.1828
p=0.5884
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

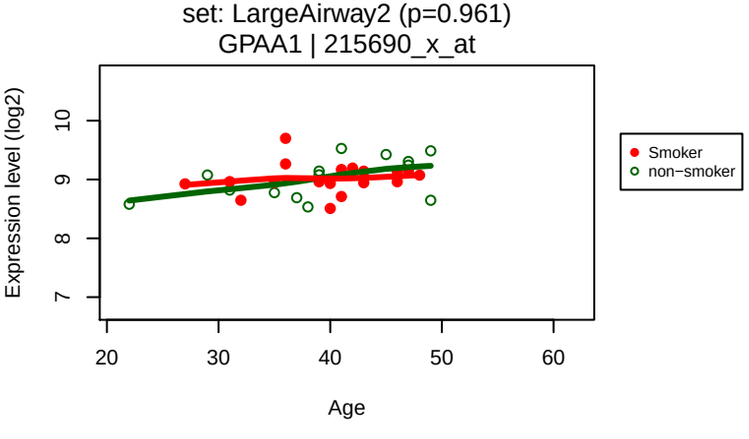

Gene expression ratio (smokers/nonsmokers)
gene: GPAA1 | 215690_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6333
p=0.944
p=0.6736
p=0.0883
p=0.961
p=0.9369
p=0.1828
p=0.5884
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

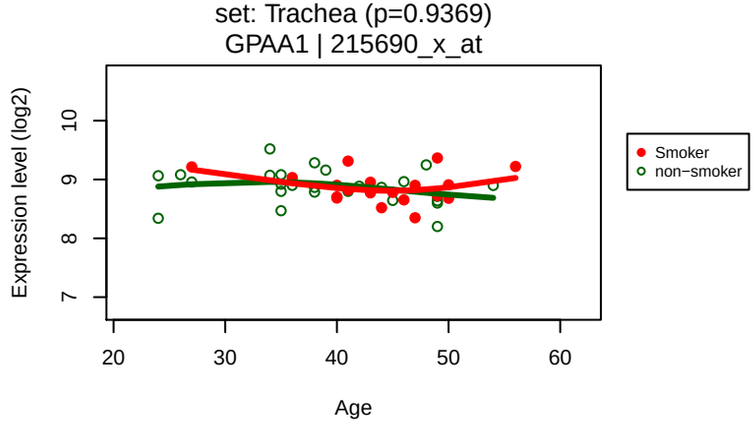

Gene expression ratio in Males
gene: GPAA1 | 215690_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6772
p=0.3159
p=0.6147
p=0.5574
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPAA1 | 215690_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1831
p=0.1903
p=0.183
p=0.7964
SmallAirway2
LargeAirway2
Trachea
Macrophage2

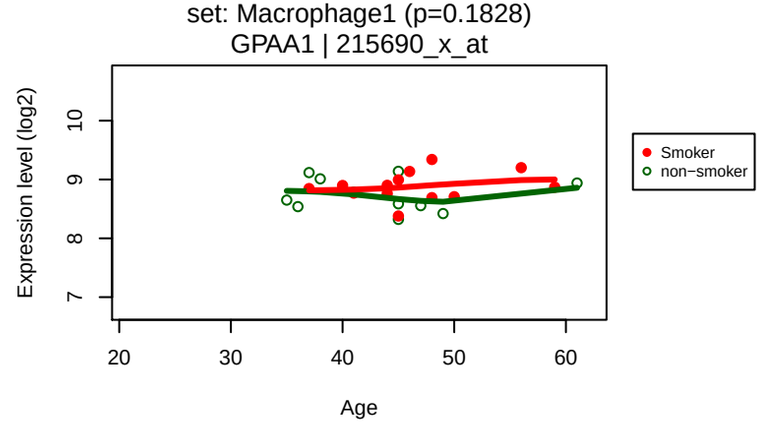

set: SmallAirway1 (p=0.6333)
GPAA1 | 215690_x_at
Expression level (log2)
Age
Smoker
non−smoker

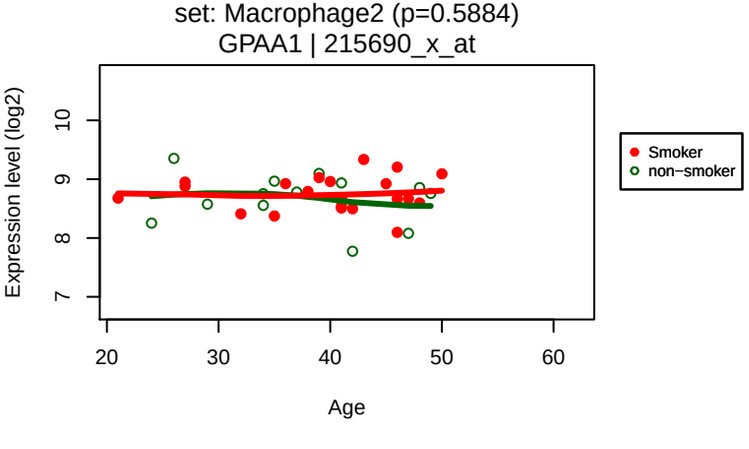

set: SmallAirway2 (p=0.944)
GPAA1 | 215690_x_at
Expression level (log2)
Age
Smoker
non−smoker

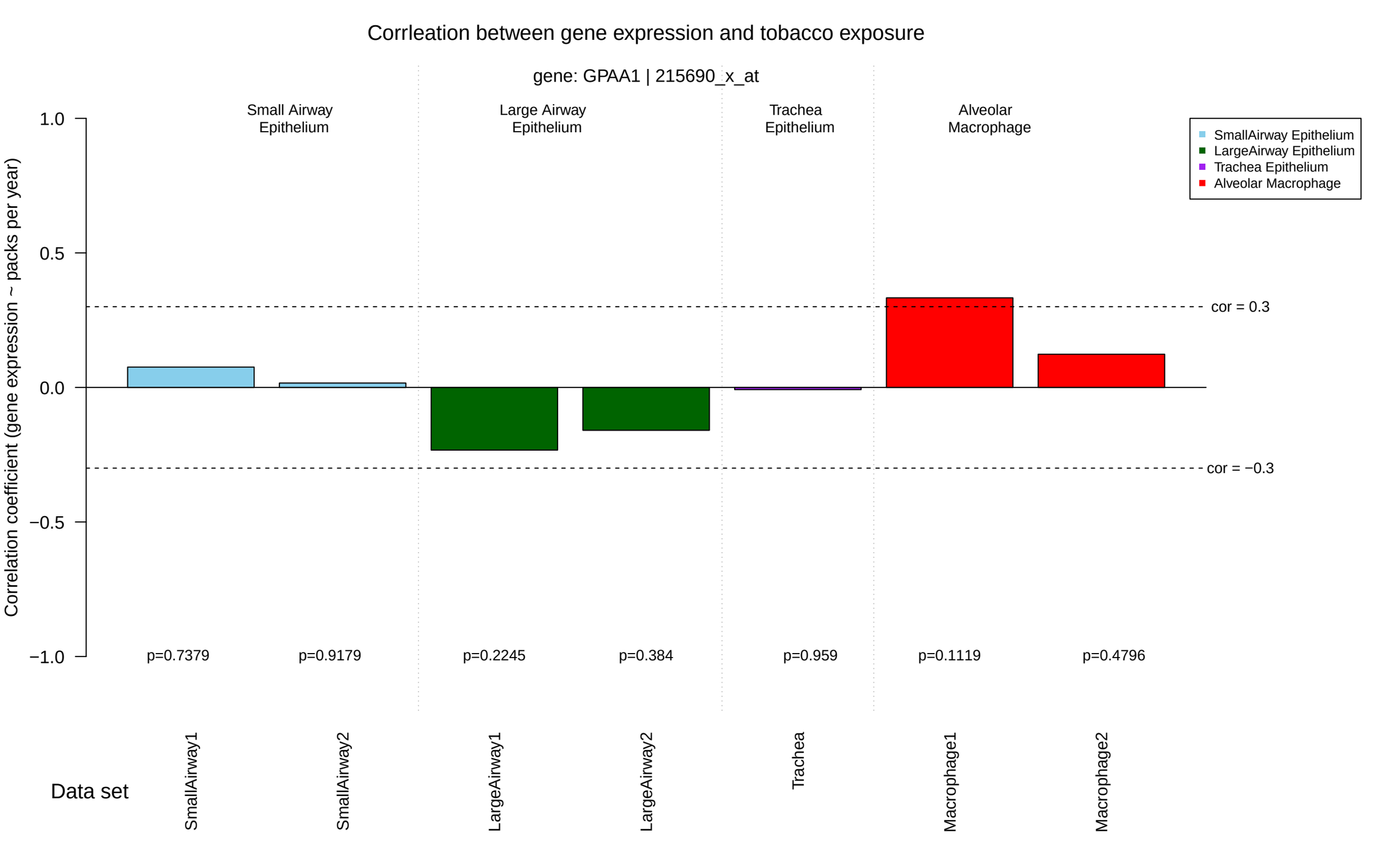

set: LargeAirway1 (p=0.0883)
GPAA1 | 215690_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.961)
GPAA1 | 215690_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9369)
GPAA1 | 215690_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1828)
GPAA1 | 215690_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5884)
GPAA1 | 215690_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPAA1 | 215690_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7379
p=0.9179
p=0.2245
p=0.384
p=0.959
p=0.1119
p=0.4796
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2